Customer Behavior Analysis Using E-Commerce Data

Exploratory Data Analysis and Descriptive Statistics

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Data Science / Data Analytics Project

2026

Introduction

Understanding customer purchasing behavior is essential for businesses operating in e-commerce platforms. Analyzing user activity, product purchases, and order patterns can provide valuable insights that help businesses improve marketing strategies, optimize product offerings, and increase customer engagement. This analysis explores an e-commerce dataset consisting of users, orders, order items, products, and payments to better understand how customers interact with the platform.

The goal of this stage of the analysis is to perform an initial exploration of the dataset using descriptive statistics. These statistics help summarize the key characteristics of the data and reveal early patterns in customer behavior and purchasing activity. By examining metrics such as the number of users, number of orders, total revenue, and product purchase distribution, we can begin to understand how customers engage with the platform and how products perform.

Descriptive statistics were calculated using SQL queries that combine multiple tables in the dataset. Aggregation functions such as COUNT, SUM, and AVG were used to summarize the data, while JOIN operations were applied to connect related information across tables including Users, Orders, Order_Items, Products, and Payments. This exploratory analysis provides a foundation for identifying trends, validating initial assumptions, and guiding deeper analytical investigations in later stages of the project.

The insights generated from this exploration are intended for business stakeholders such as product managers, marketing teams, and business analysts who are interested in understanding customer behavior and product performance within the platform.

INITIAL OBSERVATIONS AND KEY FINDINGS

The exploratory analysis of the dataset revealed several early patterns in customer behavior and purchasing activity. One of the key observations is that a subset of users appears to place multiple orders, indicating the presence of repeat customers who may play an important role in overall platform activity. This suggests that customer retention could be a significant factor influencing business performance.

Another observation is that product purchases are not evenly distributed across all products. Certain products appear more frequently in order records, which may indicate higher demand or popularity within specific categories. This uneven distribution highlights the importance of identifying top-performing products and understanding customer preferences.

Additionally, the structure of order data suggests that many orders include multiple items rather than single-product purchases. This behavior can have a direct impact on the average order value and overall revenue generation. Understanding how customers bundle products together can help businesses improve recommendations and cross-selling strategies.

These initial findings provide a starting point for deeper analysis and help identify areas that require further investigation in order to better understand customer behavior and product performance.
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EVALUATION OF INITIAL HYPOTHESES

At the beginning of the project, several assumptions were made regarding customer behavior and product performance within the platform. One of the key hypotheses was that a relatively small group of highly active users contributes a significant portion of the total revenue. Based on the initial descriptive analysis, this assumption appears to be partially supported, as some users are observed placing multiple orders. However, further analysis is required to quantify their exact contribution to overall revenue
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Another hypothesis suggested that certain product categories generate higher sales compared to others. Early observations indicate that some products are purchased more frequently, which may reflect differences in customer demand. While this provides an initial indication of product popularity, additional aggregation and category-level analysis will be necessary to confirm whether specific categories consistently outperform others in terms of revenue.

A third assumption was that repeat customers contribute more consistently to platform activity than one-time buyers. Preliminary findings show patterns of repeated purchasing behavior among certain users, but more detailed analysis will be required to measure customer retention and long-term engagement.

At this stage, the hypotheses remain plausible but not fully confirmed. Further analytical queries and deeper exploration in subsequent stages of the project will be required to validate these assumptions and draw more definitive conclusions.

ADDITIONAL QUESTIONS FOR FURTHER ANALYSIS

The exploratory analysis of the dataset highlights several areas that require deeper investigation to better understand customer behavior and business performance. One important question is identifying which customers generate the highest lifetime value and whether repeat customers contribute disproportionately to total revenue. Understanding this can help businesses focus on retaining high-value users and improving long-term engagement.

Another area of interest is analyzing product performance at a more detailed level. It would be useful to determine which product categories generate the most revenue and whether certain products are frequently purchased together. This type of analysis can support better product recommendations, bundling strategies, and inventory planning.

Additionally, examining purchasing patterns over time could provide insights into trends or seasonal variations in customer behavior. Analyzing how order frequency and revenue change across different time periods may help businesses make more informed decisions related to marketing campaigns and demand forecasting.

These questions will guide the next phase of analysis, where more advanced SQL queries and deeper data exploration will be used to uncover meaningful patterns and actionable business insights.